

Final exam of Statistics & Probability (2^h30)

Name :

Groupe:

Exercice 1. A researcher collected data on the daily commute times (in minutes) for 50 employees at a company. The data is as follows:

25, 30, 30, 35, 40, 45, 50, 50, 50, 55, 60, 65, 70, 70, 70, 75, 80, 85, 90, 90, 95, 100, 100, 100, 105, 110, 115, 120, 120, 125, 130, 135, 140, 140, 145, 150, 150, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210

1. Calculate the following measures

(a) – Mean

$$\bar{X} = \frac{1}{50} \sum_{i=1}^{50} x_i = \frac{5585}{50} = 111.7 \quad (0.5)$$

(b) – Median

We have $N = 50$ (even), therefore

$$\tilde{X} = \frac{1}{2} \left(X_{(\frac{N}{2})} + X_{(\frac{N}{2}+1)} \right) = \frac{1}{2} (X_{(25)} + X_{(26)}) = \frac{1}{2} (105 + 110) = 107.5 \quad (0.25)$$

(c) – Mode

The data is multimodal, meaning it has several modes. Each mode occurs with a frequency of 3, and the modes $\hat{X}$ are $\{50, 70, 100, 150\}$. (0.5)

(d) – Range

$$R = 210 - 25 = 185 \quad (0.25)$$

(e) – Variance and Standard deviation

Since the total number of employees is unknown, we typically assume that data collected from a sample and we have

$$Var(X) = \frac{1}{N-1} \underbrace{\sum_{i=1}^N (x_i - \bar{X})^2}_{=137830.5} = \frac{N}{N-1} \left(\underbrace{\overline{X^2}}_{=15233.5} - \bar{X}^2 \right) = 2812.867 \quad (0.75)$$

and

$$STD(X) = \sqrt{VAR(X)} = \sqrt{2812.867} = 53.036$$

(f) – Interquartile Range (IQR)

first need to find the first quartile (Q_1) and the third quartile (Q_3). The IQR is then the difference between these two values.

We have $(\frac{N+1}{4}) = 12.75 \notin \mathbb{N}$ and $3 \times (\frac{N+1}{4}) = 38.25 \notin \mathbb{N}$, then

$$Q_1 = \frac{1}{2} (X_{(12)} + X_{(13)}) = 67.5 \quad \text{and} \quad Q_3 = \frac{1}{2} (X_{(38)} + X_{(39)}) = 152.5$$

thus

$$IQR = Q_3 - Q_1 = 85 \quad (1)$$

2. Is the data symmetric, positively skewed, or negatively skewed? Justify your answer

The data exhibits a slight positive skew as indicated by Pearson's coefficient of skewness

$$sk = \frac{3(\bar{X} - \tilde{X})}{s} = \frac{3(111.7 - 107.5)}{53.036} = 0.23757 \quad (0.5)$$

3. Are there any potential outliers? (Use the 1.5 x IQR rule to check)

The lower and upper thresholds are defined by

$$Q_1 - 1.5 \times IQR = -60 \quad \text{and} \quad Q_3 + 1.5 \times IQR = 280. \quad (0.5)$$

Since all data points fall within the range of $[-60, 280]$, there are no outliers

4. Create a frequency distribution table (first, give the number of classes).

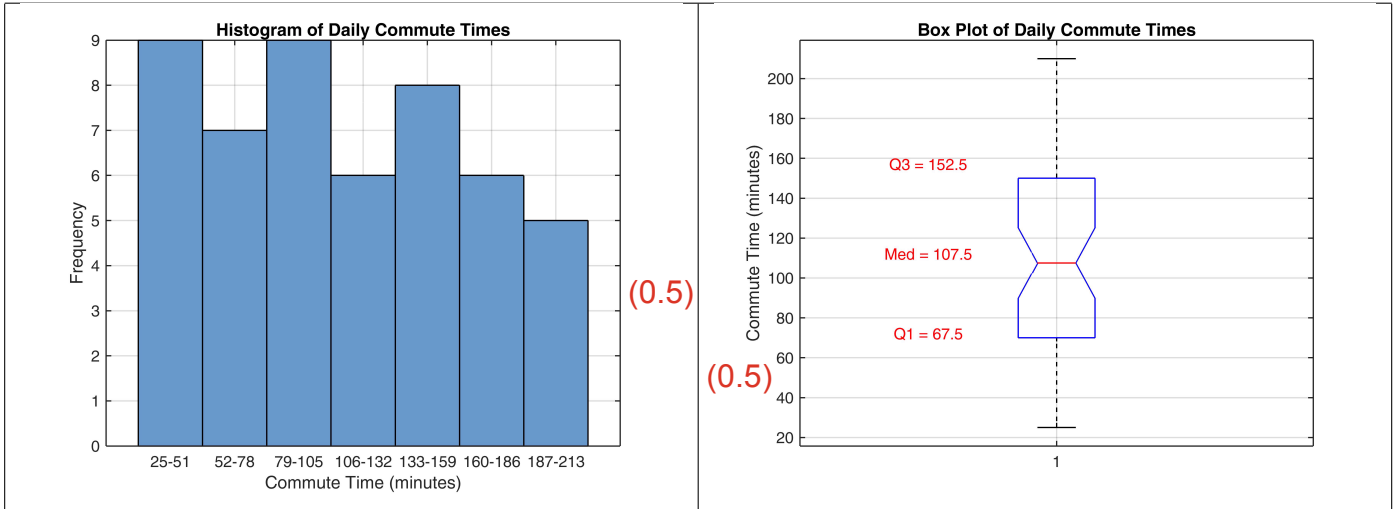
We give first the number of classes $K = \sqrt{N} = \sqrt{50} \approx 7$. Since the range $R = 185$, then we can take the width $w = \frac{R}{K} = \frac{185}{7} \approx 27$. (0.5)

Now, we create the frequency distribution table. The classes start at the minimum value (25) and increase by the class width (27):

Class Interval	25 – 51	52 – 78	79 – 105	106 – 132	133 – 159	160 – 186	187 – 213
Frequency	9	7	9	6	8	6	5

(0.75)

5. Draw a histogram based on the frequency distribution and construct a box plot.



6. If the researcher removes the top 10% of the data (the longest commute times), how does this affect the mean and standard deviation? Recalculate these measures after removing the top 10%.

First, we need to identify the top 10% of the data. We have $9 \times \left(\frac{N+1}{10}\right) = 45.9 \notin \mathbb{N}$, then

$$D_9 = \frac{1}{2} (X_{(45)} + X_{(46)}) = 187.5 \quad (0.5)$$

so the top 10% corresponds to the 5 longest commute times $\{190, 195, 200, 205, 210\}$.

Removing the top 10% of the data reduces both the mean and the standard deviation, as the largest values (which contribute significantly to both measures) are eliminated.

Therefore

$$\bar{X}^* = \frac{1}{45} [50\bar{X} - (190 + 195 + 200 + 205 + 210)] = 101.89$$

and

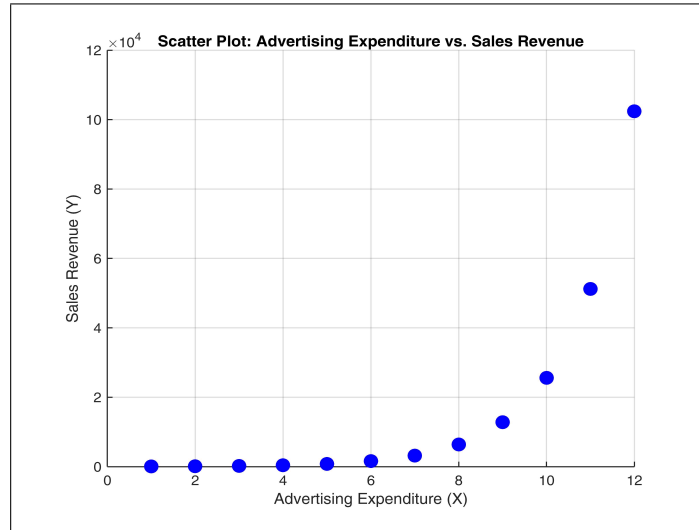
$$\overline{X^2}^* = \frac{1}{45} [50\overline{X^2} - (190^2 + 195^2 + 200^2 + 205^2 + 210^2)] = 12476 \quad (1)$$

$$\text{hence } Var(X)^* = \frac{N-5}{(N-5)-1} \left(\overline{X^2}^* - (\bar{X}^*)^2 \right) = 2142.374 \quad \text{and} \quad STD(X)^* = 46.286$$

Exercise 2. A company collected data on its monthly advertising expenditure (X) (in hundreds of dollars) and sales revenue (Y) (in dollars) for the past 12 months. The data is provided below.

Month	1	2	3	4	5	6	7	8	9	10	11	12
X	5	2	8	1	10	4	7	3	12	6	9	11
Y	800	100	6400	50	25600	400	3200	200	102400	1600	12800	51200
$Z = f(Y)$	6.685	4.605	8.764	3.912	10.150	5.991	8.071	5.298	11.537	7.378	9.457	10.843

1. Create a scatter plot of Advertising Expenditure (X) vs. Sales Revenue (Y) and comment on the visual relationship between the two variables. Is it linear or non-linear? (0.5)



The scatter plot of Advertising Expenditure (X) vs. Sales Revenue (Y) shows a non-linear relationship. The data points do not fall along a straight line. Instead, they exhibit a curved pattern. As X (Advertising Expenditure) increases, Y (Sales Revenue) increases at an accelerating rate. This is characteristic of an exponential relationship. (0.5)

2. Assume that the relationship between X and Y is exponential, i.e.

$$Y = be^{aX}$$

Apply an appropriate transformation $Z = f(Y)$ to make the relationship linear than complete the new variable in the above table.

Since the relationship is exponential, we take the natural logarithm ($\ln$) of the dependent variable Y , then

$$\ln(Y) = aX + \ln(b) \quad (0.25)$$

3. What is the appropriate correlation coefficient to measure the relationship between Advertising Expenditure (X) and Sales Revenue (Y) before any transformation? Compute this correlation coefficient (between X and Y) and interpret its value.

Before any transformation, the relationship between Advertising Expenditure (X) and Sales Revenue (Y) is non-linear (exponential). In such case, the Spearman rank correlation coefficient (r_s) is the appropriate measure.

$$r_s = \frac{Cov(R(X), R(Y))}{\sigma_{R(X)}\sigma_{R(Y)}} = 1 - \frac{6 \sum_{i=1}^n d_i^2}{N(N-1)} \quad (1)$$

where $d_i = R(X_i) - R(Y_i)$, and we have $\sum_{i=1}^n d_i^2 = 0$; therefore $r_s = 1$ which indicates a perfect positive monotonic relationship between X and Y . As Advertising Expenditure (X) increases, Sales Revenue (Y) tends to increase in a consistent (but non-linear) manner.

4. What is the appropriate correlation coefficient to measure the relationship between Advertising Expenditure (X) and the transformed Sales Revenue ($Z = f(Y)$) after applying the transformation? Compute this correlation coefficient (between X and Z) and interpret its value.

After applying the transformation $Z = \ln(Y)$, the relationship between Advertising Expenditure (X) and Transformed Sales Revenue (Z) becomes linear. In this case, the Pearson correlation coefficient (ρ) is the appropriate measure.

$$\rho(X, Z) = \frac{Cov(X, Z)}{\sigma_X \sigma_Z} = \frac{\overline{XZ} - \overline{X} \times \overline{Z}}{\sqrt{(\overline{X^2} - \overline{X}^2)(\overline{Z^2} - \overline{Z}^2)}}$$

we have

$$\begin{aligned} \overline{X} &= \frac{1}{12} \times 78 = 6.5, \overline{X^2} = \frac{1}{12} \times 650 = 54.167, \\ \overline{Z} &= \frac{1}{12} \times 92.692 = 7.724, \overline{Z^2} = \frac{1}{12} \times 784.688 = 65.391, \\ \overline{XZ} &= \frac{1}{12} \times 701.618 = 58.468 \end{aligned} \quad (1)$$

thus,

$$\rho(X, Z) = 0.99952$$

This indicates a almost perfect positive linear relationship between X and Z . As Advertising Expenditure (X) increases, the Transformed Sales Revenue ($Z = \ln(Y)$) tends to increase linearly.

5. Write the regression equation of Z on X and interpret the slope and intercept in the context of the problem.

We have

$$Z = \ln(Y) = aX + b' \text{ where } b' = \ln(b)$$

and a and b' are given by

$$\begin{aligned} a &= \frac{Cov(X, Z)}{Var(X)} = \frac{8.262}{11.917} = 0.693 \\ \text{and} \\ b' &= \overline{Z} - a\overline{X} = 3.2195 \end{aligned} \quad (1)$$

then

$$Z = 0.693X + 3.2195$$

For every additional \$100 spent on advertising, $\ln(Y)$ increases by $a = 0.693$. If no money is spent on advertising, $\ln(Y) = b' = 3.912$. (0.25)

6. Compute the residual variance and the variance explained by the regression line $D_{Z|X}$. Do you think that the regression line fit well the data, measure it.

the residual variance :

$$\begin{aligned} V_R(Z) &= Var(Z) (1 - \rho(X, Z)^2) \\ &= 5.730 (1 - 0.99952^2) \\ &= 5.4995 \times 10^{-3} \end{aligned} \quad (0.25)$$

the explained variance

$$V_E(Z) = Var(Z) - V_R(Z) \stackrel{or}{=} Var(Z) \rho(X, Z)^2 = 5.7245 \quad (0.25)$$

To determine how well the regression line fits the data, we use the coefficient of determination (R^2). It is calculated as:

$$R^2 = \frac{V_E(Z)}{Var(Z)} = \rho(X, Z)^2 = 0.99904 \quad (0.5)$$

it indicates that 99.9% of the variability in Z is explained by the regression line. This suggests that the regression line fits the data very well.

7. Reverse the transformation to express the regression equation in terms of the original variables and use the regression equation to predict the sales revenue for: Advertising expenditure = 13 (hundreds of dollars).

if $\ln(Y) = aX + b' = 0.693X + 3.2195$ then

$$Y = e^{0.693X + 3.2195} = 25.016 \exp(0.693X) \quad (0.25)$$

To predict the sales revenue for an advertising expenditure of $X = 13$ (hundreds of dollars), substitute $X = 13$ into the regression equation

$$\hat{Y} = 25.016 \exp(0.693 \times 13) = 2.0454 \times 10^5 \text{ dollars} \quad (0.25)$$

Exercise 3. A company has three factories (A, B, and C) that produce light bulbs. The production rates and defect rates for each factory are as follows:

- Factory A: Produces 40% of the total light bulbs, with a 2% defect rate.
- Factory B: Produces 35% of the total light bulbs, with a 3% defect rate.
- Factory C: Produces 25% of the total light bulbs, with a 5% defect rate.

A light bulb is randomly selected from the company's inventory.

1. (a) – What is the probability that the selected light bulb was produced by Factory A?

$$\text{Set } A = \text{the selected light bulb was produced by Factory A} \implies P(A) = 0.4 \quad (0.25)$$

- (b) – What is the probability that the selected light bulb was produced by Factory B?

Set B = the selected light bulb was produced by Factory B $\implies P(B) = 0.35$ (0.25)

(c) – What is the probability that the selected light bulb was produced by Factory C?

Set C = the selected light bulb was produced by Factory C $\implies P(C) = 0.25$ (0.25)

2. What is the probability that the selected light bulb is defective?

The overall probability of a bulb being defective (D) is calculated using the law of total probability:

$$P(D) = P(D | A) \cdot P(A) + P(D | B) \cdot P(B) + P(D | C) \cdot P(C).$$

From the problem, we have

$$P(D | A) = 0.02, P(D | B) = 0.03, P(D | C) = 0.05 \quad (1)$$

Substitute these values :

$$P(D) = (0.02 \times 0.40) + (0.03 \times 0.35) + (0.05 \times 0.25) = 0.031$$

3. Given that the selected light bulb is defective, what is the probability that it was produced by Factory A?

$$P(A | D) = \frac{P(A \cap D)}{P(D)} = \frac{P(D | A) \cdot P(A)}{P(D)} = \frac{0.02 \times 0.40}{0.031} = 0.25806 \quad (0.75)$$

4. Given that the selected light bulb is defective, what is the probability that it was produced by Factory B or C?

$$\begin{aligned} P(B \cup C | D) &= \frac{P((B \cup C) \cap D)}{P(D)} = \frac{P((B \cap D) \cup (C \cap D))}{P(D)} = \frac{P(B \cap D) + P(C \cap D)}{P(D)} \quad (1) \\ &= \frac{P(D | B) \cdot P(B) + P(D | C) \cdot P(C)}{P(D)} = \frac{(0.03 \times 0.35) + (0.05 \times 0.25)}{0.031} = 0.74194 \end{aligned}$$

5. Assume that we select six light bulbs one by one at random, What is the probability that at least two consecutive selected light bulbs are defective?

Instead of directly calculating the probability of at least two consecutive defective bulbs, we calculate the probability of the complementary event (No two consecutive defective bulbs) and subtract it from 1.

Set p = Probability of a bulb being defective (D) = 0.031.

$q = 1 - p$: Probability of a bulb being non-defective (N) = $1 - 0.031 = 0.969$.

Let A_n be the valid sequences of length n with no two consecutive defective bulbs. The base cases are

$$\begin{aligned} A_1 &= \text{either defective (D) or non-defective (N)} \\ A_2 &= \{NN, ND, DN\} \end{aligned}$$

We have $P(A_1) = p + q = 1$ and $P(A_2) = q^2 + qp + pq = q^2 + 2pq = 0.99904$

The recurrence relation for A_n is:

$$P(A_n) = P(A_{n-1}) \cdot q + P(A_{n-2}) \cdot p \cdot q$$

Where $P(A_{n-1}) \cdot q$ represents the probability of A_{n-1} adding the last bulb is not defective and $P(A_{n-2}) \cdot p \cdot q$ is the probability of A_{n-2} adding the last bulb is defective, but the second-to-last bulb is not defective.

Here, we are looking for $P(A_6)$ solution of

$$P(A_6) = P(A_5) \cdot q + P(A_4) \cdot p \cdot q$$

we have $P(A_3) = P(A_2) \cdot q + P(A_1) \cdot p \cdot q = 0.99904 \times 0.969 + 0.031 \times 0.969 = 0.99811$, similarly, we can find $P(A_4) = 0.99717$, $P(A_5) = 0.99624$, and

$$P(A_6) = 0.99624 \times 0.969 + 0.99717 \times 0.031 \times 0.969 = 0.99531$$

(2)

Finally, the probability that at least two consecutive selected light bulbs are defective is

$$1 - P(A_6) = 1 - 0.99531 = 0.00469$$

Exercise 4. Let $X = \{x_1, x_2, \dots, x_n\}$ be a dataset with sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$. Define the k^{th} sample central moments m_k as:

$$m_k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^k$$

Prove the following inequality between the fourth central moment m_4 and the second central moment m_2 :

$$m_4 \geq m_2^2.$$

Solution Using Cauchy-Schwarz: The Cauchy-Schwarz inequality for sums states that for any two sequences $\{u_i\}$ and $\{v_i\}$:

$$\left(\sum_{i=1}^n u_i^2 \right) \left(\sum_{i=1}^n v_i^2 \right) \geq \left(\sum_{i=1}^n u_i v_i \right)^2$$

To prove $m_4 \geq m_2^2$, let's define:

$$u_i = (x_i - \bar{X})^2 \quad \text{and} \quad v_i = 1$$

Substituting into the Cauchy-Schwarz inequality:

(2)

$$\left(\sum_{i=1}^n (x_i - \bar{X})^4 \right) \underbrace{\left(\sum_{i=1}^n 1^2 \right)}_{=n} \geq \left(\sum_{i=1}^n (x_i - \bar{X})^2 \right)^2$$

then divide both sides by n^2

$$m_4 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^4 \geq \left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2 \right)^2 = m_2^2.$$

Solution using Jensen's inequality: Using Jensen's inequality for the convex function $\phi(x) = x^2$, we have

$$\frac{1}{n} \sum_{i=1}^n \phi((x_i - \bar{X})^2) \geq \phi\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2\right)$$

therefore

$$m_4 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^4 \geq \left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2 \right)^2 = m_2^2$$